

Kinetic Mechanism of Beef Heart Ubiquinol:Cytochrome *c* Oxidoreductase¹

Tomomi Kubota,* Shinya Yoshikawa,** and Hiroshi Matsubara*

*Department of Biology, Faculty of Science, Osaka University, Toyonaka, Osaka 560; and **Department of Life Science, Faculty of Science, Himeji Institute of Technology, Harima Science Park, Kamigori, Ako-gun, Hyogo 678-12

Received for publication, July 29, 1991

The electron transfer from ubiquinol-2 to ferricytochrome *c* mediated by ubiquinol:cytochrome *c* oxidoreductase [E.C. 1.10.2.2] purified from beef heart mitochondria, which contained one equivalent of ubiquinone-10 (Q_{10}), was investigated under initial steady-state conditions. The Q_{10} -depleted enzyme was as active as the Q_{10} -containing one. Double reciprocal plots for the initial steady-state rate *versus* one of the two substrates at various fixed levels of the other substrate gave parallel straight lines in the absence of any product. Intersecting straight lines were obtained in the presence of a constant level of one of the products, ferrocytochrome *c*. The other product, ubiquinone-2, did not show any significant effect on the enzymic reaction. Ferrocytochrome *c* non-competitively inhibited the enzymic reaction against either ubiquinol-2 or ferricytochrome *c*. These results indicate a Hexa-Uni ping-pong mechanism with one ubiquinol-2 and two ferricytochrome *c* molecules as the substrates, which involves the irreversible release of ubiquinone-2 as the first product and the irreversible isomerization between the release of the first ferrocytochrome *c* and the binding of the second ferricytochrome *c*. Considering the cyclic electron transfer reaction mechanism, this scheme suggests that the binding of quinone or quinol to the enzyme and electron transfer between the iron-sulfur center and cytochrome c_1 are rigorously controlled by the electron distribution within the enzyme.

Mitochondrial ubiquinol:ferricytochrome *c* oxidoreductase is the middle component of the respiratory chain, which catalyzes the transfer of electrons from ubiquinol to cytochrome *c*, coupled with proton translocation across the inner mitochondrial membrane (1, 2). Extensive studies have been performed on this enzyme because of its important role in energy coupling during oxidative phosphorylation at site 2. This enzyme and others analogous to it have been isolated from various organisms (3). All the enzymes contain two *b*-type and one *c*-type cytochromes and an iron-sulfur protein. The beef heart enzyme has been most thoroughly studied, and contains 11 different subunits (4). A gross three-dimensional structure of a dimeric nature has been proposed for the *Neurospora crassa* enzyme on the basis of the results of an electron microscopic study (5). However, no X-ray crystallographic structure has been reported, although the crystallization of the enzyme from beef heart has been successfully accomplished (6, 7). The redox behavior of the purified enzyme, as well as the membrane bound form, has been studied extensively using various inhibitors (1). The results obtained, especially as to the oxidant-induced reduction of cytochrome *b* in the presence of antimycin-A (8), strongly suggest a cyclic electron transfer reaction within the enzyme molecule (2). Several mechanisms have been proposed for this electron transfer within the molecule, coupled with proton trans-

location (9-12).

It has been proven that extensive analysis of the initial steady-state kinetics for an enzyme system involving multiple substrates and products provides highly unique information about the reaction mechanism, such as the order of substrate binding and product release, and the number of intermediate enzyme species (13). However, only a few preliminary reports on the initial steady-state kinetics for this enzyme have appeared, and the conclusions drawn have been inconsistent, for example, a ping-pong mechanism (14) *versus* a sequential one (15), and linear Lineweaver-Burk plots (16) *versus* non-linear ones (17).

This paper deals with the steady-state kinetics for the beef heart enzyme system involving ubiquinol-2 (Q_2H_2) and ferricytochrome *c* (cytochrome c^{3+}) as the substrates. The present results indicate Hexa-Uni ping-pong mechanism involving a strong repulsive interaction between the two substrate binding sites and the irreversible release of one of the products, ubiquinone-2 (Q_2).

MATERIALS AND METHODS

Q_2 was kindly donated by Eisai, Tokyo. Q_2H_2 was prepared by the method of Rieske (18). Horse heart cytochrome *c* was a product of Sigma (Type VI). The enzyme was prepared from fresh beef heart muscle by the method of Rieske (18) with slight modifications, as follows. A Keilin-Hartree preparation (19) instead of a mitochondrial preparation was used as the starting material, the protein concentration of which was increased to 35 from 25 mg/ml. The Keilin-Hartree preparation was made up within 4 h

¹ This work was supported in part by Grants-in-Aid for Scientific Research on Priority Areas of Bioenergetics, and grant No. 01470148 from the Ministry of Education, Science and Culture of Japan. Abbreviations: Q_2 , ubiquinone-2; Q_2H_2 , ubiquinol-2; Q_{10} , ubiquinone-10; $Q_{10}H_2$, ubiquinol-10.

after slaughter. The freshness of the beef heart seems critical for the consistency of the preparation. The enzyme preparation was stored in an ice bath at about 200 μ M and used within 3 weeks without any detectable change in enzymic activity.

The enzyme concentration was expressed in terms of the cytochrome c_1 concentration determined from the absorbance difference for the fully reduced form (dithionite-reduced form) between 554 and 540 nm minus that for the fully oxidized form, using a difference extinction coefficient of 19 $\text{mM}^{-1}\cdot\text{cm}^{-1}$ (20). The concentrations of cytochrome c and Q_2H_2 were estimated using $\Delta\epsilon_{550\text{nm}}^{\text{red-ox}}$ of 18.5 $\text{mM}^{-1}\cdot\text{cm}^{-1}$ and $\Delta\epsilon_{290\text{nm}}$ of 4.14 $\text{mM}^{-1}\cdot\text{cm}^{-1}$ in ethanol (21), respectively. The content of ubiquinone-10 (Q_{10}) in the enzyme preparation was determined by the method of Redfearn (22).

The Q_{10} -depleted cytochrome bc_1 complex was prepared as follows. The stock enzyme solution was diluted to about 20 μ M (5 mg/ml) with a buffer comprising 50 mM Tris-HCl (pH 8.0), 660 mM sucrose, 1 mM L-histidine, and 0.1% (w/v) sucrose monolaurate. The enzyme was precipitated by the addition of polyethylene glycol-4000 [30% (w/v)] and then collected by centrifugation. The precipitate was then dissolved and washed with the same buffer as used for the dilution, except that it contained 0.5% sucrose monolaurate, in an ultrafiltration apparatus with an ADVANTEC UK-200 membrane. The resulting enzyme preparation, the Q_{10} -depleted cytochrome bc_1 complex, showed the same absorption spectrum and SDS-PAGE pattern as before the treatment (7). This preparation contained about 0.05 equivalents of Q_{10} per cytochrome c_1 . Reconstitution of Q_{10} was carried out by the method reported by Yu *et al.* (23).

The temperature of the reaction mixture in the cell was maintained at $20 \pm 0.1^\circ\text{C}$ in a thermostated water-jacketed cell holder with a small magnetic stirrer. The rate of electron transfer from Q_2H_2 to cytochrome c^{3+} catalyzed by the enzyme was estimated from the rate of reduction of cytochrome c measured at 550 nm in a buffer comprising 55 mM Tris-HCl (pH 7.4), 2 mM NaN_3 , 0.5 mM Na-EDTA, 0.01% Tween-20, and 20 mM MgCl_2 , unless otherwise stated. Non-enzymic electron transfer between Q_2H_2 and cytochrome c was estimated by measuring the linear increase in absorbance at 550 nm for 1 min in the absence of the enzyme. The rate of enzymic reduction was monitored for 2 min after the addition of the enzyme. Then, a slight excess of solid dithionite was added to determine the absorbance level at the fully reduced state of cytochrome c . The enzymic reaction rate was estimated by subtracting the non-enzymic reduction, which was less than 1% of the initial rate of the enzymic reaction under the present assay conditions. The enzyme solution for each assay was made up daily by dilution of the stock solution (about 200 μ M) to 1 μ M with a buffer comprising 50 mM Tris-HCl (pH 8.0), 660 mM sucrose, 1 mM L-histidine, and 0.01% Tween-20, and kept in an ice-bath. The enzymic assay measurements were performed within 3 h using the diluted enzyme solution after incubation for 2 h at 0°C . The specific activity remained constant for the 3-h period. The initial rate data were fitted to the Michaelis-Menten equation with a non-linear least square regression program, kindly written by Prof. T. Tsukihara. The enzymic activity was expressed in terms of the turnover number, s^{-1} (mol of cytochrome c

reduced/ cytochrome c_1/s).

RESULTS

Enzyme Preparation and Conditions for the Enzyme Assay—The present modification of the preparation method significantly improved the reproducibility as to both the yield and the enzymic activity. The yield was 500 ± 100 mg per 800 g of minced beef heart muscle. The specific activity was 360 $\mu\text{mol}/\text{min}/\text{mg}$ protein. This preparation contains 1.1 equivalents of Q_{10} to cytochrome c_1 .

Incubation of the enzyme solution in the presence of Tween-20 at 0°C for 2 h prior to the reaction, careful temperature control of the reaction mixture and the presence of Tween-20 in the reaction medium, as stated above, were most effective for improving the experimental accuracy. The experimental accuracy was 1.2% ($\pm 4.6 \text{ s}^{-1}/393 \text{ s}^{-1}$) for the system containing Q_2H_2 (25 μM), cytochrome c^{3+} (10 μM), and the enzyme (600 pM), but 10% ($\pm 21 \text{ s}^{-1}/210 \text{ s}^{-1}$) in the absence of the detergent at room temperature using the enzyme without preincubation. The variance of the specific activity depending on the preparation batch was $\pm 6\%$ ($\pm 11 \text{ s}^{-1}/180 \text{ s}^{-1}$) with 10 μM Q_2H_2 and 10 μM cytochrome c^{3+} . The set of data given in each figure in this paper was obtained with using a single batch of the enzyme preparation. The variance of the enzymic activity in each preparation was not large enough to affect the initial rate pattern.

Stoichiometry—The stoichiometry of Q_2H_2 oxidized versus cytochrome c^{3+} reduced was determined in order to confirm that the observed reduction of cytochrome c^{3+} was due to the electron transfer reaction catalyzed by the enzyme. The assay system contained various concentrations of cytochrome c^{3+} and a constant concentration of Q_2H_2 , 10 μM . The enzyme concentration (5 nM) was higher than that under the usual assay conditions in order to obtain rapid completion of the reaction. The amount of cytochrome c reduced during the reaction was plotted against the total concentration of cytochrome c in the reaction mixture, as shown in Fig. 1. Cytochrome c^{3+} contained initially in the system was completely reduced as long as the cytochrome c concentration was lower than 23 μM . Above that concentration, only 23 μM of the cytochrome c^{3+} initially present was reduced. This result indicates that Q_2H_2 is oxidized stoichiometrically by cytochrome c^{3+} . The enzymic reduction was completely inhibited by antimycin-A (5 μM).

Time Course of the Enzymic Reaction—The enzymic reduction of cytochrome c proceeded as shown typically in Fig. 2. Each point represents the time-dependent decrease in the fractional concentration of cytochrome c^{3+} (ratio of [cytochrome c^{3+}] during the reaction to the initial [cytochrome c^{3+}] level). The initial concentrations of cytochrome c^{3+} , Q_2H_2 , and the enzyme were 7.6, 15 μM , and 0.6 nM for the open triangles, and 20.7, 19 μM , and 1.0 nM for the open circles, respectively. The solid curves are for the first order reaction with respect to the cytochrome c^{3+} concentration. Slight but systematic deviations from the solid curves were obvious from 60 s after initiation of the reaction. The deviations in other determinations performed thus far were smaller than those in the upper trace in Fig. 2. The initial velocity was estimated from the first-order rate constant multiplied by the initial concentration

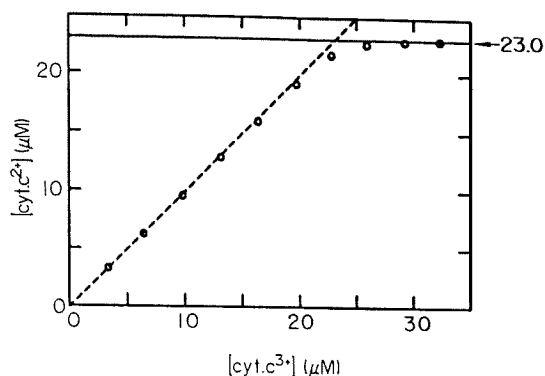


Fig. 1. Stoichiometry of the reaction between Q_2H_2 and cytochrome c^{3+} . The amounts of cytochrome c reduced were plotted against the total concentration of cytochrome c^{3+} at a constant level of $10 \mu M$ Q_2H_2 . The reaction catalyzed by 5 nM enzyme was completed within 5 min . The arrow indicates $23.0 \mu M$, the estimated maximum concentration of cytochrome c reduced with Q_2H_2 in the system.

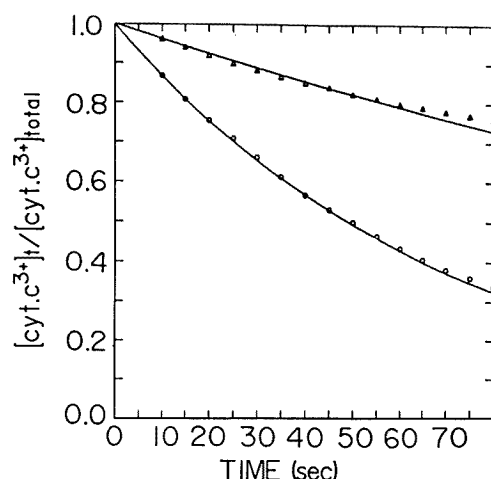


Fig. 2. The time course of the electron transfer from Q_2H_2 to cytochrome c^{3+} catalyzed by the enzyme. The ratios of the cytochrome c concentration at each time to the initial concentration were plotted against time after initiation of the reaction. Non-enzymic reduction of cytochrome c^{3+} was subtracted from the data. The initial concentrations of cytochrome c , Q_2H_2 , and the enzyme were 7.6 , $15 \mu M$, and 0.6 nM for open triangles, and 20.7 , $19 \mu M$, and 1.0 nM for open circles, respectively. The solid curves are the best-fit exponential curves with first-order rate constants of $4.0 \times 10^{-3} \text{ s}^{-1}$ for the open triangles and $14.2 \times 10^{-3} \text{ s}^{-1}$ for the open circles. The final level of absorbance to be obtained after completion of the reaction was estimated from the absorbance level obtained on addition of a slight excess of dithionite to the reaction mixture.

of cytochrome c^{3+} . The initial rate thus obtained was proportional to the enzyme concentration at any fixed initial concentrations of the two substrates. Therefore, the enzyme concentration was adjusted to provide a reaction rate appropriate for accurate measurement under the present experimental conditions ($t_{1/2}$; $30\text{--}300 \text{ s}$).

The K_m value for cytochrome c^{3+} was decreased from 4 to $1 \mu M$ without affecting V_{max} by eliminating 20 mM $MgCl_2$ from the reaction mixture, while K_m for Q_2H_2 was not effected by $MgCl_2$ (data not shown). These results suggest that $MgCl_2$ affects the binding of cytochrome c^{3+} to the

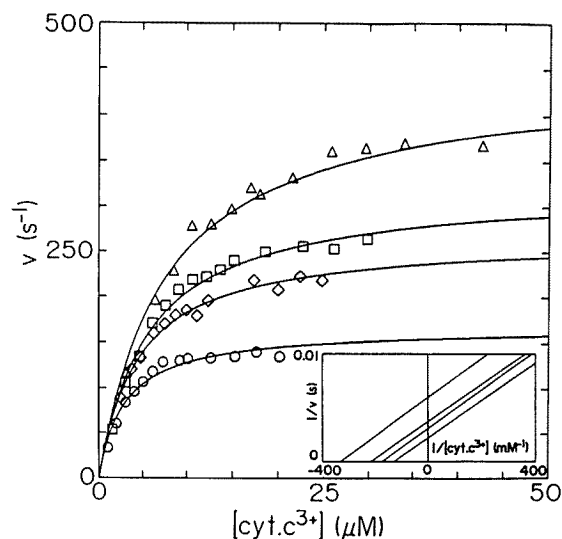


Fig. 3. Dependence of the initial rate on the concentration of cytochrome c^{3+} at various fixed levels of Q_2H_2 . Q_2H_2 was fixed at various levels, as follows: triangles, $25 \mu M$; squares, $15 \mu M$; diamonds, $10 \mu M$; circles, $7 \mu M$. Each point shows the initial rate obtained from the first-order rate constant multiplied by the concentration of cytochrome c^{3+} . The best-fit curves based on the Michaelis-Menten equation are also given. The Lineweaver-Burk plots in the inset were obtained from the parameters given in Table I.

TABLE I. Kinetic parameters for ferricytochrome c with various levels of Q_2H_2 . The data were obtained with the Michaelis-Menten equation using a non-linear least-square regression program.

Concentration of Q_2H_2 (μM)	K_m (μM)	V_{max} (s^{-1})	K_m/V_{max} (nMs)
25	7.7 ± 0.71	450 ± 12	17 ± 1.7
15	5.7 ± 0.49	322 ± 8.9	18 ± 1.6
10	4.6 ± 0.35	266 ± 6.4	17 ± 1.4
7	3.0 ± 0.41	167 ± 6.4	18 ± 2.5

enzyme without affecting any other part of the reaction mechanism. In the present study, $MgCl_2$ was added to the reaction mixture because the K_m value in the absence of $MgCl_2$ was too small for accurate determination under the present experimental conditions.

Q_{10} -Depleted Cytochrome bc_1 Complex—The enzymic activity of the Q_{10} -depleted cytochrome bc_1 complex was measured in the same reaction medium as above except that it contained 0.05% sucrose-monolaurate instead of 0.01% Tween-20. Under these conditions, the activity of the Q_{10} -depleted enzyme was 370 s^{-1} in the system containing $10 \mu M$ of both substrates, while those for the Q_{10} -reconstituted enzyme and the enzyme as prepared, which contained 1.1 equivalent of Q_{10} as described above, were 380 and 350 s^{-1} , respectively. These activity values are comparable to the reported value (24). This showed that Q_{10} was not a prerequisite for the enzymic activity, which conflicted with the observation of Yu *et al.* (23). This discrepancy seems to be due to the difference in the procedures for Q_{10} -depletion; only one precipitation with polyethylene glycol-4000 in the presence of sucrose monolaurate versus several precipitations with ammonium sulfate in the presence of cholate (23). The former procedure is milder than the latter, considering the ionic strength

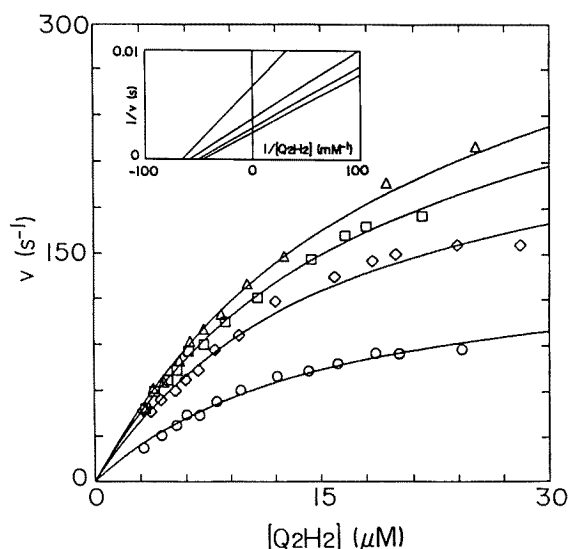


Fig. 4. Dependence of the initial rate on the concentration of Q_2H_2 in the presence of cytochrome c^{3+} . All measurements were carried out in an assay mixture containing a constant level of $10 \mu M$ cytochrome c^{3+} . The cytochrome c^{3+} level was fixed at various levels, as follows: triangles, $30 \mu M$; squares, $20 \mu M$; diamonds, $12 \mu M$; circles, $7 \mu M$. Each point shows the initial rate value vs. the concentration of Q_2H_2 . Solid curves are the best-fit curves obtained as in Fig. 4. The Lineweaver-Burk plots in the inset were obtained using the parameters given in Table II.

sensitivity of this enzyme (25). It is unlikely that the Q_{10} attached to the enzyme mediates electron transfer from Q_2H_2 to the enzyme, as has been suggested.

Initial Steady-State Rate Pattern—The dependency of the initial steady-state rate as to the concentration of cytochrome c^{3+} was examined with various levels of Q_2H_2 . According to Cleland's notation (13), cytochrome c^{3+} and Q_2H_2 are a variable substrate and a changing fixed substrate, respectively. As shown in Fig. 3, at each level of Q_2H_2 , the initial rate pattern was a rectangular hyperbolic curve with respect to the substrate concentration. The solid lines are best-fit curves computed as described above. No systematic deviation from the experimental curves was observed. The computed values of K_m and V_{max} are given in Table I. The slopes of the Lineweaver-Burk plots, K_m/V_{max} , shown in the table are independent of the Q_2H_2 level within the limits of experimental accuracy, giving a series of parallel Lineweaver-Burk lines, as shown in the inset in Fig. 3. The initial rate pattern for Q_2H_2 and cytochrome c^{3+} as the variable substrate and the changing fixed substrate, respectively, estimated from the solid lines in Fig. 3, provides a set of linear and parallel Lineweaver-Burk plots. This pattern was confirmed experimentally (data not shown). Therefore, the reciprocal rate is a linear function with respect to the reciprocal concentrations of both substrates, as follows,

$$1/v = K_1/A + K_2/B + 1/V_{max}$$

where A and B are the substrate concentrations, and K_1 , K_2 , and V_{max} are the constants independent of A and B . This result indicates ping-pong-type binding and release of the substrates and products.

Effects of Products on the Initial Steady-State Rate

TABLE II. Kinetic parameters for Q_2H_2 with various levels of ferricytochrome c in the presence of ferrocytochrome c . The data were obtained as in Table I.

Concentration of ferricytochrome c (μM)	K_m (μM)	V_{max} (s^{-1})	K_m/V_{max} (nMs)
30	21 ± 3.0	400 ± 32	53 ± 8.7
20	19 ± 1.3	340 ± 13	56 ± 4.3
12	17 ± 1.7	260 ± 14	63 ± 7.2
7	15 ± 1.6	149 ± 8.5	100 ± 12

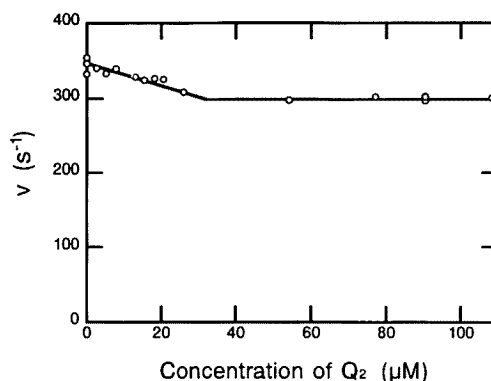


Fig. 5. Effect of Q_2 on the initial rate. Initial rates were measured at $30 \mu M$ Q_2H_2 and $25 \mu M$ cytochrome c^{3+} in the presence of various concentrations of Q_2 . A constant volume of ethanol, $10 \mu l$, containing a constant amount of Q_2H_2 and various amounts of Q_2 was added to the reaction mixture.

Pattern—The effect of a product at a constant level on the initial velocity pattern was examined in order to obtain further evidence for the ping-pong mechanism. The initial rate pattern for the enzyme system containing Q_2H_2 and cytochrome c^{3+} as the variable and changing fixed substrates, respectively, in the presence of a constant level of ferrocytochrome c (cytochrome c^{2+}), $10 \mu M$, as shown in Fig. 4, gave a series of Michaelis-Menten patterns with the K_m and V_{max} values given in Table II. The computed values of K_m/V_{max} in Table II depend on the level of cytochrome c^{3+} , giving the intersecting double reciprocal plots shown in the inset in Fig. 4. The intersecting pattern in the presence of a product is one of the most important pieces of evidence for a ping-pong mechanism. On the other hand, the other product, Q_2 , up to $100 \mu M$, showed only a 20% decrease in activity, as shown in Fig. 5. An increase in Q_2 concentration above $50 \mu M$ did not affect the enzymic activity. Therefore, the small decrease in activity induced by Q_2 up to $40 \mu M$ is not due to product inhibition caused by the direct competition of Q_2 with one of the substrates. Therefore, this result (absence of product inhibition) indicates the irreversible release of Q_2 from the enzyme under turnover conditions.

Inhibition Patterns of a Product, Cytochrome c^{2+} —The product inhibition pattern of cytochrome c^{2+} was examined in order to prove the involvement of more than two molecules of cytochrome c^{3+} as a substrate in the enzyme system. The inhibitory effect of cytochrome c^{2+} on the initial rate pattern for the enzyme system with Q_2H_2 as the variable substrate at a fixed concentration of $10 \mu M$ cytochrome c^{3+} was examined, as shown in Fig. 6. The results clearly indicated that cytochrome c^{2+} inhibits the enzymic reaction non-competitively with respect to Q_2H_2 . This non-competitive inhibitory effect is considered to be due to

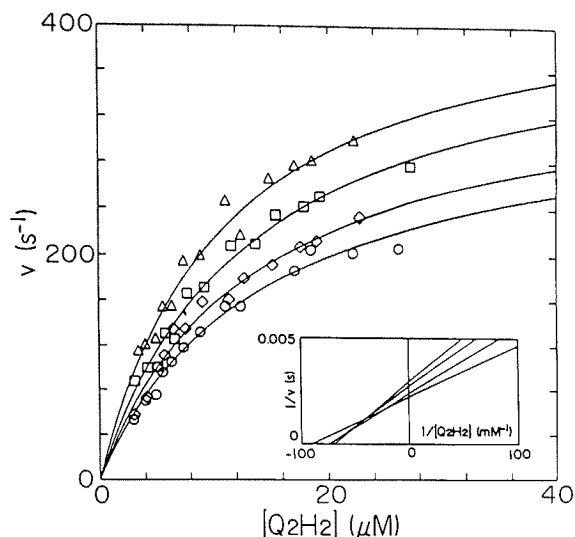


Fig. 6. Effect of cytochrome c^{2+} on the initial rate of the reaction system with Q_2H_2 as a variable substrate. The concentration of cytochrome c^{3+} was fixed at $10 \mu M$. The cytochrome c^{2+} concentration was fixed at various levels, as follows: triangles, $0 \mu M$; squares, $5 \mu M$; diamonds, $10 \mu M$; circles, $15 \mu M$. The initial rate is plotted against the concentration of Q_2H_2 . The best-fit curves are also given. The inset shows Lineweaver-Burk plots calculated using the parameters given in Table III.

TABLE III. Kinetic parameters for Q_2H_2 as to the effect of product inhibition of ferrocyanochrome c with $10 \mu M$ ferrocyanochrome c . The data were obtained as in Table I.

Concentration of ferrocyanochrome c (μM)	K_m (μM)	V_{max} (s^{-1})	K_m/V_{max} (nMs)
0	11 ± 1.4	450 ± 27	26 ± 3.4
5	13 ± 1.5	420 ± 24	32 ± 3.9
10	14 ± 1.6	370 ± 23	37 ± 4.4
15	14 ± 1.9	340 ± 24	42 ± 6.3

the non-competitive inhibitory effect of the cytochrome c^{2+} released as the second product against the first binding substrate, Q_2H_2 , although cytochrome c^{2+} released as the last product competes for E (Scheme 1) with Q_2H_2 . As shown in the inset in Fig. 6, the double reciprocal patterns calculated from the best-fit curve to the experimental results in this figure (see Table III for kinetic parameters) seem to have points of intersection that depend on the concentration of cytochrome c^{2+} . However, statistical examination of the experimental accuracy indicated no significant difference in the positions of these intersection points. The effect of cytochrome c^{2+} on the initial rate pattern was examined for a system containing cytochrome c^{3+} as the variable substrate and Q_2H_2 ($20 \mu M$) as the fixed substrate. As shown in the inset in Fig. 7, cytochrome c^{2+} inhibited the enzyme reaction with a tendency of non-competitive inhibition against cytochrome c^{3+} . This figure preferably indicates that no systematic deviation from Michaelis-Menten kinetics was detected for the initial rate pattern in the presence of cytochrome c^{2+} , suggesting an irreversible step between the release of cytochrome c^{2+} as the second product and the binding of cytochrome c^{3+} as the third substrate.

The simplest mechanism derived from all of the above

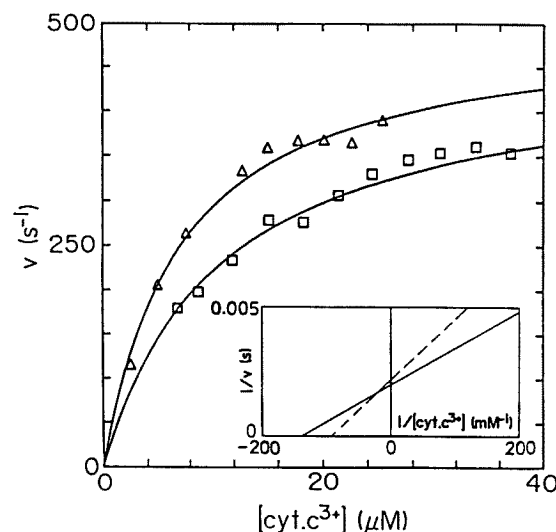
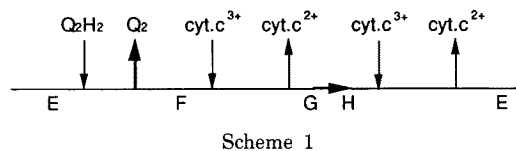


Fig. 7. Effect of cytochrome c^{2+} on the initial rate with cytochrome c^{3+} as a variable substrate. Initial rates were measured at a fixed level of $20 \mu M$ Q_2H_2 and various levels of cytochrome c^{3+} in the absence or presence of cytochrome c^{2+} ; triangles, $0 \mu M$; squares, $10 \mu M$. Each symbol shows the initial rate vs. the concentration of cytochrome c^{3+} . The Lineweaver-Burk plots in the inset were obtained using K_m and V_{max} values of $7.17 \mu M$ and $502 s^{-1}$ for the solid line, and $10.8 \mu M$ and $459.6 s^{-1}$ for the broken line, respectively.



results is a Hexa-Uni ping-pong mechanism in terms of Cleland's notation, as shown in Scheme 1, where the release of Q_2 and the isomerization of G to H represented by the bold arrows, are irreversible.

DISCUSSION

Deligi Esposti and Lenaz (15) reported that the initial steady-state rate of the beef heart enzyme in the absence of any products gave intersecting double reciprocal plots suggesting a sequential mechanism. However, they determined this by fitting straight lines to the double reciprocal plots assuming constant reliability for each point of the reciprocal rate, regardless of the variable substrate concentration. As is well known, such analysis is likely to produce a misleading conclusion. Kauten *et al.* (14) reported, without giving any experimental results, that the yeast enzyme showed ping-pong-type kinetics. Non-linear double reciprocal plots for quinol or quinol derivatives as variable substrates have been reported for the enzyme in its isolated (16) as well as the membrane-bound form (15). However, the non-linearity was proposed based on double reciprocal plots without any statistical analysis. Therefore, more extensive analyses of the initial steady-state kinetics, with careful consideration to experimental accuracy, are required in order to establish the steady-state properties of this enzyme system.

Non-linear least-square fitting of the Michaelis-Menten equation to the present experimental results was quite successful, as shown in Figs. 3, 4, 6, and 7. No systematic deviation from the equation was detectable. Scheme 1 is the simplest mechanism explaining the present results, and indicates that the second substrate, cytochrome c^{3+} , reacts with the two-electron-reduced form after the first product, Q, has been released from the enzyme. This suggests a strong negative cooperativity between the Q_2H_2 and cytochrome c^{3+} binding sites, since the fairly hydrophobic molecule, Q_2H_2 , is unlikely to compete for a single site with a highly hydrophilic protein, cytochrome c^{3+} .

The Q_{10} -depleted cytochrome bc_1 complex showed comparable enzymic activity to that of the untreated enzyme, which contained one equivalent of Q_{10} per cytochrome c_1 . This result suggests that Q_{10} is not a prerequisite for the enzymic activity, that is, Q_2H_2 in a bulk solution functions as a direct substrate (a quinol-pool). Perhaps Q_{10} produced on the oxidation of ubiquinol-10 ($Q_{10}H_2$) at the active site is released irreversibly from the ubiquinone binding site like Q_2 . However, the Q_{10} released from the active site is likely to stay on the enzyme surface because of its hydrophobicity. Thus, this result strongly suggests that Q_2H_2 binds to the same site as that which accepts $Q_{10}H_2$. In other words, the electron transfer reaction within the enzyme after it has received electrons from the substrate is likely to be identical regardless of the quinol structure, $Q_{10}H_2$, in the membrane or, Q_2H_2 , in a solution.

As is well known, two mechanisms, the Q-cycle (double-

turnover mechanism) (11) and the semiquinone-cycle (9), have been proposed for interpretation of the cyclic electron transfer reactions within the enzyme, as given schematically in Fig. 8. The present steady-state kinetic results can be used to evaluate these proposals and provide new insights in to the reaction mechanism.

The absence of any significant effect of Q_2 on the enzyme reaction under turnover conditions (Fig. 5) strongly conflicts with the Q-cycle mechanism (Fig. 8A), since in this mechanism, Q_2 must bind to the enzyme as a substrate to accomplish the catalytic cycle. Thus, the present results support the semiquinone-cycle rather than the Q-cycle. One of the simplest schemes which explains the present kinetic results as well as the semiquinone-cycle mechanism is shown in Scheme 2, where the bold arrows denote irreversible reaction steps and the regular arrows, reversible steps. The possible distribution of reducing equivalents in each enzyme form is also given in the scheme. Each substrate (or product)-bound form is not given for the sake of simplicity. The rate equation for this scheme is given in the Appendix. The irreversible release of Q_2 was proven experimentally in the present study. Reversible isomerization from G to H is required for the non-competitive inhibition of cytochrome c^{2+} against Q_2H_2 , as shown in Fig. 6. If the second Q_2H_2 binds to G without forming H, cytochrome c^{2+} would compete with Q_2H_2 for G as well as E, leading to competitive inhibition against Q_2H_2 . The release of Q_2H_2 must be irreversible in order to provide the parallel double reciprocal plots in the absence of any product (Fig. 3) and to give linear double reciprocal plots for the initial rate *versus* the concentration of cytochrome c^{3+} as the variable substrate in the presence of a constant level of cytochrome c^{2+} (Fig. 7). This means that the affinity of Q_2H_2 is rigorously controlled by the electron distribution within the enzyme, as in the case of the enzyme species, H *versus* I. The second Q_2H_2 to bind to H causes an electron left on the low potential centers from the first Q_2H_2 to migrate to the high potential centers.

Furthermore, another unique scheme was deduced from the semiquinone-cycle, as shown in Scheme 3, which does not include the binding of two molecules of Q_2H_2 under turnover conditions, but requires initial activation of the enzyme from E to G. In this mechanism, the enzyme form, G, in which one molecule of cytochrome b is reduced, is an active enzyme form that receives electrons from Q_2H_2 .

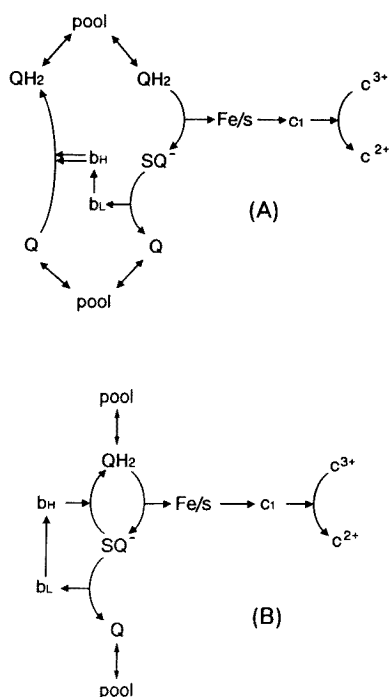
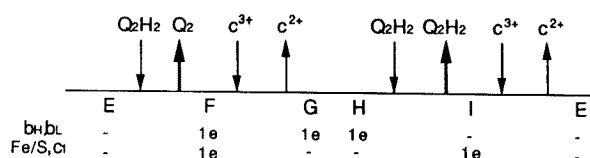
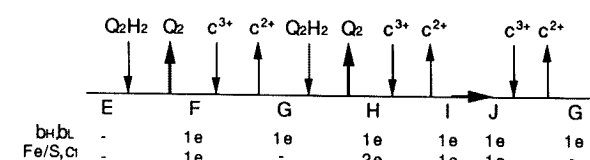


Fig. 8. The Q-cycle (A) and the semiquinone-cycle (B) mechanism for electron transfer through the bc_1 complex. The Q-cycle (double turnover mechanism) was modified from Crofts *et al.* (11). The semiquinone-cycle proposed by Wikström and Krab (9) is shown without the concept of center- i and center- o . b_L , low-potential heme b_{668} ; b_H , high-potential heme b_{682} ; c_1 , cytochrome c_1 ; c , cytochrome c ; Fe/S, Rieske iron-sulfur center; pool, ubiquinone pool in the membrane.



Scheme 2



Scheme 3

under turnover conditions (see the Appendix for the rate equation). The fully oxidized enzyme, E, and the two-electron-reduced species, F, do not participate in the catalytic cycle. In the presteady-state, G is formed from E *via* F. No kinetic information on the initial activation can be obtained from the results of initial steady-state analysis. The formation of H from G may be described as follows; Q₂H₂ bound to G only reduces Fe/S, forming the semiquinoid form of Q₂ (ubisemiquinone-2). Then, this ubisemiquinone-2 is reduced to Q₂H₂ with an electron from one of the cytochromes *b*. The second formed Q₂H₂, without release, reduces the enzyme form in which one of the high potential redox centers is reduced, giving H in which one electron equivalent is at one of the low potential centers and two equivalents are at the high potential centers, as shown in the scheme. This mechanism also requires a strong control of Q₂H₂ binding by the oxidation state of the enzyme, that is, one electron equivalent at the high potential centers in the intermediate producing the second formed Q₂H₂ prevents the release of the Q₂H₂ from the enzyme. The isomerization of I to J in Scheme 3 is required for linear double reciprocal plots for the initial rate *versus* the concentration of cytochrome *c*³⁺ as the variable substrate in the presence of a constant level of cytochrome *c*²⁺ (Fig. 7). Tsai *et al.* (26) reported a rapid equilibrium in the electron transfer reaction between Fe/S and cytochrome *c*₁, based on the results of stopped-flow kinetic analysis of the ferricyanide oxidation of the ascorbate-reduced enzyme in which only cytochrome *c*₁ and the Fe/S cluster were reduced. Preliminary stopped flow analysis of the reaction between cytochrome *c*²⁺ and the fully oxidized enzyme (to be published elsewhere) confirmed the reversible electron transfer between the two redox centers. The irreversible isomerization of I to J in this scheme, which possibly corresponds to the electron transfer from the Fe/S site to cytochrome *c*₁, apparently conflicts with these stopped flow data. However, one electron equivalent is at one of the cytochromes *b* in I and J, while both cytochromes *b* were oxidized in the enzyme species in the above stopped-flow experiments. Therefore, the reverse electron transfer (isomerization of J to I) is likely to be prevented by the reduced cytochrome *b*.

In conclusion, the simplest schemes which explain the experimental results obtained so far are Schemes 2 and 3, which indicate rigorous mechanisms of control through the electron distribution within the enzyme molecule for the binding of ubiquinone or ubiquinol to the enzyme (in Schemes 2 and 3), and for the electron transfer reaction

between the Fe/S cluster and cytochrome *c*₁ (in Scheme 3). Both schemes involve a strong repulsive interaction between the two substrate binding sites, for ubiquinone and cytochrome *c*.

REFERENCES

1. Hatefi, Y. (1985) *Annu. Rev. Biochem.* **54**, 1015-1069
2. Rieske, J.S. (1976) *Biochim. Biophys. Acta* **456**, 195-247
3. Trumpower, B.L. (1990) *Microbiol. Rev.* **54**, 101-129
4. González-Halphen, D., Lindorfer, M.A., & Capaldi, R.A. (1988) *Biochemistry* **27**, 7021-7031
5. Li, Y., De Vries, S., Leonard, K., & Weiss, H. (1981) *FEBS Lett.* **135**, 277-280
6. Yue, W.-H., Zou, Y.-P., Yu, L., & Yu, C.-A. (1991) *Biochemistry* **30**, 2303-2306
7. Kubota, T., Kawamoto, M., Fukuyama, K., Shinzawa-Itoh, K., Yoshikawa, S., & Matsubara, H. (1991) *J. Mol. Biol.* **221**, 379-382
8. Rieske, J.S. (1971) *Arch. Biochem. Biophys.* **145**, 179-193
9. Wikström, M. & Krab, K. (1986) *J. Bioenerg. Biomembr.* **18**, 181-193
10. Mitchell, P. (1976) *J. Theor. Biol.* **62**, 327-367
11. Crofts, A.R., Meinhardt, S.W., Jones, K.R., & Snozzi, M. (1983) *Biochim. Biophys. Acta* **723**, 202-218
12. Slater, E.C. (1983) *Trends Biochem. Sci.* **8**, 239-242
13. Cleland, W.W. (1963) *Biochim. Biophys. Acta* **67**, 104-137
14. Kauten, R., Tsai, A.-L., & Palmer, G. (1987) *J. Biol. Chem.* **262**, 8658-8667
15. Deligi Esposti, M. & Lenaz, G. (1982) *Biochim. Biophys. Acta* **682**, 189-200
16. Nalecz, M.J. & Azzi, A. (1985) *Arch. Biochem. Biophys.* **240**, 921-931
17. Deligi Esposti, M. & Lenaz, G. (1982) *Arch. Biochem. Biophys.* **216**, 727-735
18. Rieske, J.S. (1967) *Methods Enzymol.* **10**, 239-245
19. Yoshikawa, S., Choc, M.G., O'Toole, M.C., & Caughey, W.S. (1977) *J. Biol. Chem.* **252**, 5498-5508
20. Hatefi, Y., Haavik, A.G., & Griffiths, D.E. (1962) *J. Biol. Chem.* **237**, 1681-1685
21. Morrison, L.E., Schelhorn, J., E., Cotton, T.M., Bering, C.L., & Loach, P.A. (1982) in *Function of Quinones in an Energy Conserving System* (Trumpower, B.E., ed.) pp. 35-58, Academic Press, New York
22. Redfearn, E.R. (1967) *Methods Enzymol.* **10**, 381-384
23. Yu, L., Yu, C.-A., & King, T.E. (1978) *J. Biol. Chem.* **253**, 2657-2663
24. Deligi Esposti, M., Tsai, A.-L., Palmer, G., & Lenaz, G. (1986) *Eur. J. Biochem.* **160**, 547-555
25. Rieske, J.S., Baum, H., Stoner, C.D., & Lipton, H. (1967) *J. Biol. Chem.* **242**, 4854-4866
26. Tsai, A.-L., Olson, S., & Palmer, G. (1983) *J. Biol. Chem.* **258**, 2122-2125

Supplemental Materials

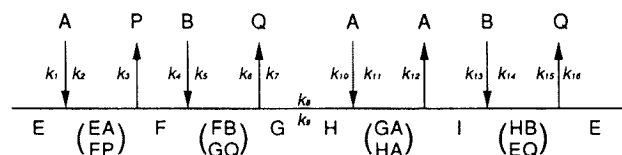
Appendix

Each scheme is presented according to Cleland's notation [13];

A, ubiquinol-2; B, ferricytochrome c; P, ubiquinone -2; Q,

ferricytochrome c; E, F, G, H, I and J, enzyme species; k_n , rate constants.

Scheme 2:



Rate equation:

$$\frac{Et}{v} = \left(K_1 \frac{1}{AB} + K_2 \frac{1}{A} + K_3 \frac{1}{B} + K_4 \right) Q + K_5 \frac{1}{A} + K_6 \frac{1}{B} + K_7$$

where the constants, K_n , are defined as follows:

$$K_1 = \frac{k_2 k_{14} k_{16} + k_3 k_{14} k_{18}}{k_1 k_3 k_{12} k_{15}} + \frac{k_5 k_7 k_9 k_{11} + k_5 k_7 k_{10} k_{12}}{k_4 k_6 k_8 k_{10} k_{12}}$$

$$K_2 = \frac{k_2 k_9 k_{12} + k_7 k_9 k_{11}}{k_4 k_6 k_{10} k_{12}} + \frac{k_2 k_{18} + k_3 k_{18}}{k_1 k_3 k_{15}}$$

$$K_3 = \frac{k_5 k_7}{k_4 k_6 k_8}$$

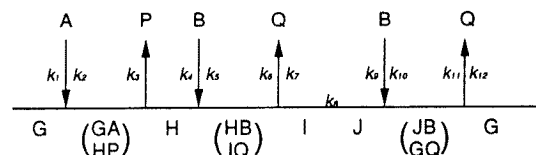
$$K_4 = \frac{k_7}{k_6 k_8}$$

$$K_5 = \frac{k_2 k_{11} + k_3 k_{11}}{k_4 k_{10} k_{12}} + \frac{k_2 + k_3}{k_1 k_3} + \frac{1}{k_{10}}$$

$$K_6 = \frac{k_3 + k_6}{k_4 k_6} + \frac{k_{14} + k_{18}}{k_{12} k_{15}} + \frac{k_9}{k_4 k_{10}}$$

$$K_7 = \frac{1}{k_3} + \frac{1}{k_6} + \frac{1}{k_8} + \frac{1}{k_{12}} + \frac{1}{k_{15}}$$

Scheme 3:



Rate equation:

$$\frac{Et}{v} = \left(K_1 \frac{1}{AB} + K_2 \frac{1}{A} + K_3 \frac{1}{B} + K_4 \right) Q + K_5 \frac{1}{A} + K_6 \frac{1}{B} + K_7$$

where A, B, P and Q are the concentrations of the reactants represented

by the corresponding letters, and K_n are constants defined as follows:

$$K_1 = \frac{k_2 k_{10} k_{12} + k_3 k_{10} k_{12}}{k_1 k_3 k_6 k_{11}}$$

$$K_2 = \frac{k_2 k_{12} + k_3 k_{12}}{k_1 k_3 k_{11}}$$

$$K_3 = \frac{k_5 k_7}{k_4 k_6 k_8}$$

$$K_4 = \frac{k_7}{k_6 k_8}$$

$$K_5 = \frac{k_2 + k_3}{k_1 k_3}$$

$$K_6 = \frac{k_3 + k_6}{k_4 k_6} + \frac{k_{10} + k_{11}}{k_6 k_{11}}$$

$$K_7 = \frac{1}{k_3} + \frac{1}{k_6} + \frac{1}{k_8} + \frac{1}{k_{11}}$$